PaperSonar

Project proposal

Design baseline 1.0 / 25 October 2026

PaperSonar is a research intelligence platform for comparing research directions through publication trends, citation context and scholarly participation. The proposed first release is a reproducible PostgreSQL warehouse with a set of analytical reports and inspectable supporting records.

Problem and intended users

Researchers can retrieve many relevant papers yet still struggle to compare a few possible directions consistently. Search results do not directly explain whether a topic is growing within a defined corpus, whether citation comparisons account for publication age, or whether a proposed intersection has supporting work worth reading. Research groups also need to retain the evidence used when a direction was selected.

The intended users are researchers preparing a literature review, research supervisors comparing possible directions, and small R&D groups investigating an unfamiliar area. PaperSonar will organize the metadata needed for those comparisons. It will not certify a scientific gap or replace reading the publications.

Objectives

- Integrate a bounded set of OpenAlex publication records and reconcile selected DOI metadata with Crossref.
- Model publication output, citation observations and citation relationships at explicit and separate grains.
- Answer fourteen analytical questions through repeatable SQL and Power BI reports.
- Keep source records, corpus rules, metric definitions and dated exports so results can be traced and rebuilt.
- Support later product validation through a web interface that compares directions and exposes supporting papers.

Initial coverage

Start with selected software engineering topics and related intersections. The historical baseline covers 2016 to 2025. Data from 2026 will be reported separately as an incomplete year through the extraction cutoff. The target of 30,000 to 60,000 admitted works will be adjusted after source profiling to keep the corpus complete within its declared filters.

Approach and expected outputs

Python batch jobs will retain raw JSON and load PostgreSQL staging tables. Validation will check canonical identifiers, duplicates, unknown dimensions and complete partition loads before updating the warehouse. The dimensional model uses a publication transaction fact, a work observation snapshot fact and a factless citation relationship fact. Shared dimensions and attribution bridges support topic, time, venue, institution and author analysis.

OpenAlex supplies the primary scholarly metadata and citation relationships. Crossref supplies exact DOI reconciliation. ROR is optional institutional enrichment. A free OpenAlex key supports the bounded import, while application limits prevent paid usage [S1, S2, S8, S9]. Raw article PDFs and copyrighted abstracts are outside the data baseline.

Output	Purpose
Dimensional warehouse	Store publication events, observed citation totals and directed reference relationships.
Five report pages	Compare activity, citation cohorts, communities, potential intersections and data coverage.
Reproducible evidence	Keep raw pages, manifests, configuration, SQL and dated exports.
Documented analytical methods	State counting methods, cutoffs, denominators and limits for all fourteen questions.

Feasibility and limits

The design uses local PostgreSQL, Python and Power BI Desktop. Page-wise extraction and bounded full refreshes avoid a requirement for large-memory processing or a cloud warehouse. Citation history will start on the first successful observation date. The system cannot reconstruct past citation counts from today's total, and topic co-occurrence cannot verify an unanswered scientific problem.

Future value

The next product step is a small public application for saving topic comparisons and exporting evidence dossiers. Institutional reporting and hosted collaboration may later support a paid service. Those directions require usage tests and sustainable hosting. The first release remains useful as a reproducible bibliometric analysis toolkit.

Sources

[S1] OpenAlex API reference. https://help.openalex.org/api/

[S2] OpenAlex pricing and free daily allowance. https://help.openalex.org/access/pricing/

[S8] Crossref REST API and metadata reuse. https://www.crossref.org/documentation/retrieve-metadata/rest-api/

[S9] ROR registry releases and CC0 reuse. https://ror.readme.io/docs/data-dump